

A Bayesian network for early diagnosis of sepsis patients: a basis for a clinical decision support system

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Abstract— Sepsis is a severe medical condition caused by an inordinate immune response to an infection. Early detection of sepsis symptoms is important to prevent the progression into the more severe stages of the disease, which kills one in four it effects. Electronic medical records of 1492 patients containing 233 cases of sepsis were used in a clustering analysis to identify features that are indicative of sepsis and can be further used for training a Bayesian inference network. The Bayesian network was constructed using the systemic inflammatory response syndrome criteria, mean arterial pressure, and lactate levels for sepsis patients. The resulting network reveals a clear correlation between lactate levels and sepsis. Furthermore, it was shown that lactate levels may be predicative of the SIRS criteria. In this light, Bayesian networks of sepsis patients hold the promise of providing a clinical decision support system in the future.

Keywords: *Decision support system; CDSS; Sepsis; Severe Sepsis; Septic shock; EMR; Bayesian Network; Clustering*

I. INTRODUCTION

Sepsis is a serious burden on the medical system, especially the more dangerous forms of severe sepsis and septic shock, with an estimated \$14.6 billion spent on hospitalizations [1]. The latter two forms of the disease have a combined incidence of killing one in four that are affected [2]. Sepsis is defined by the presence of an infection and a systemic inflammatory response syndrome (SIRS) which can occur in any tissue within the body [3]. The SIRS criteria are heart rate (HR) >90 beats per minute, respiratory rate (RR) >20 breaths per minute (bpm), temperature >38°C or <36°C, white blood cell (WBC) count >12,000 or <4000 cells/mm³ or 10% bands. Any two of these criteria is sufficient to identify a patient as having SIRS. Severe sepsis is categorized as sepsis with the presence of organ dysfunction due to the sepsis itself. Septic shock is observed when there is sepsis-induced hypotension where either the systolic blood pressure (SBP) is below 90 mm Hg or the mean arterial pressure (MAP) is below 70 mm Hg [4].

Just as in stroke and acute myocardial infarction, the speed and proper therapies at the onset of severe sepsis and septic shock are necessary to improve patient outcome [4]. Progression to severe sepsis is associated with higher mortality rates and other severe conditions such as

permanent organ damage, cognitive impairment, and physical disability [1]. For this reason it is imperative to identify patients having sepsis at the onset of the symptoms before it progresses in severity.

There have been studies that have worked towards identifying clinically relevant indicators for impending septic shock, such as lactate levels [5, 6] or scores that are an aggregate of many different tests [7]. Furthermore, it was shown by Varpula et al. [8] that MAP and lactate levels were independently associated with the 30-day mortality in septic shock. These hypothesis-driven studies provide evidence of tests that can be further used in a decision support system (DSS) towards early prediction and appropriate treatments of sepsis patients (Fig. 1.1). Recently, Peelen et al. have developed a Dynamic Bayesian Network modeling the progression of organ failure based on the Sequential Organ Failure Assessment score in the intensive care unit [9] [10]. Brause et al. have used neural networks that aimed to predict the critical states of the sepsis patients, where a critical state is defined as a state where a majority of measurements are derived from deceased patients [11]. Taken together, these two DSSs demonstrate the uniqueness of our Bayesian networks to investigate relationships between tests, although both these systems do not aim to investigate the relationship between tests and outcomes.

Here, we exploit the availability of large depositories of detailed electronic medical records (EMR) in order to identify informative features. This data-assisted approach allows for the development and training of probabilistic models that can be further evaluated in a clinical setting. Toward this goal, we first analyzed an EMR database to find correlations between tests, diagnoses, and clinical outcomes. We then provide a basis for a clinical DSS that focuses on preventing sepsis progression and early treatment. To describe the associations between the various features and visualize their structure, we used a probabilistic graphical model known as a Bayesian network (BN). Bayesian networks provide a probabilistic model that can represent the joint distribution of all variables contained. Thus, an important feature of BNs is that any variable can be predicted from any independent subset of variables from within the network and a statement about the certainty of its estimation can be provided [12]. To this end, by using the values of the SIRS criteria, MAP, and lactate levels averaged

over time, with the addition of sepsis occurrence and hospitalization duration, we construct a BN to determine clinically relevant relationships between these tests and outcomes.

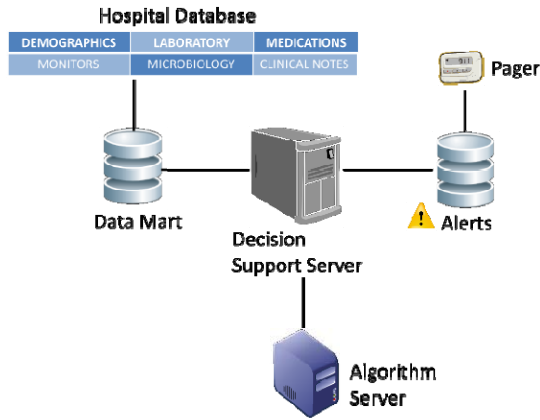


Figure 1.1: Schematics of a proposed clinical decision support system.

II. METHODS

An electronic medical records (EMR) database containing 1492 patients, with 233 separate cases of sepsis, that was provided in collaboration with the University of California Davis Medical Center was used to train and develop the BN. The SIRS criteria were included in the BN, however the heart rate test was not available in the database and therefore not included. Consequently, the following variables were used as nodes in the analysis: the five tests of temperature, RR, WBC, MAP, lactate; and the two outcomes of length of stay in the hospital (LOS) measured in days, and sepsis occurrence (SO). Analysis was performed in the R statistical software environment (<http://www.r-project.org/>) using the bnlearn package (<http://www.bnlearn.com/>).

To prepare the patients for analysis, all of the values per patient in each of the five tests were averaged over time to provide a single measure for each test administered to that patient. As a result, the time dependence of the tests was removed to simplify the construction of the network. However, not all tests were available for every patient. From 1492 patients, only 1076 of them had lactate values available and only 741 patients had the MAP measurements. Hence, the first BN was constructed for 1076, and the second for 741 patients. Furthermore, due to this reduction of patients, some of the sepsis patients were removed as well. In the first BN, only 211 cases of sepsis remained and in the second network only 151 remained.

At the outset of the analysis, a hierarchical clustering of the sepsis database was performed with the addition of the mortality patients. The goal of the clustering was to determine whether any informative groupings could be teased out of the database that may be potentially used for performing inference using the BN. For the clustering, Silhouette values were provided to show the degree of confidence in the clustering assignment. Generally, a Silhouette value of 0.50 and above demonstrates a

reasonable structure was found (>0.70 indicates strong structure) and below 0.50 a weak structure was determined; the maximum value possible is 1 and the minimum is -1. The exact number of clusters to cut the resulting tree was determined from the “elbow plot” of the within cluster dispersion as function of the number of clusters added to model the data. The cluster cutoff value of five clusters was chosen as it was the point after which adding more clusters did not affect the dispersion measure.

Building the network was a two step process. First, a network consisting of temperature, RR, WBC, lactate, LOS, and SO (BN1) was constructed. Then a second network (BN2) was constructed using the aforementioned variables with the addition of the MAP test. The purpose of performing a two-tiered analysis was to gauge the connectivity gained or lost with the addition of the MAP.

To construct the BN, the network was learned using the score-based greedy search hill-climbing algorithm, wherein the four different scoring measures of Gaussian log-likelihood (LOG), Akaike information (AIC), Bayesian information (BIC), and Bayesian Gaussian equivalent (BGE) criterion scores were used to guide the learning procedure. Score-based learning proceeds by evaluating the change in score after trying all possible arc direction changes, arc removals and arc additions, which do not produce directed loops. The change which improves the score the most is accepted after evaluating all the modifications. The procedure stops if every arc change decreases the score [12].

To determine the generalization error of our inference method and to avoid overfitting, 10-fold cross-validation on each network was performed to provide an estimate of its goodness of fit. The cross-validation performance is evaluated using a log-likelihood loss function, with lower values indicating better performance. Thus the BN with the least expected loss was chosen as the ideal network.

Also the strength of the arcs within the networks was calculated using bootstrap-based inference [13] to provide a comparison between the BN1 and BN2 performance. The directions of the arcs were determined again using the bootstrap approach with an ad-hoc direction threshold of 0.70 using the method in [14], in which a value of 1 would indicate directionality completely in that direction and 0 would represent the opposite case.

III. RESULTS

From the hierarchical clustering analysis (Fig. 3.1 and Table I, II), it was demonstrated that five clusters could be used to classify the patients. The largest clusters were clusters 3 and 4, which contained a majority of the deaths, sepsis cases, and patients. The main difference between the two clusters was that cluster 3 had a lower WBC than cluster 4, which had a WBC greater than $12,000 \text{ cells/mm}^3$ – one of the SIRS criteria.

Only cluster 5 had the characteristics that satisfied the minimum of two SIRS criteria necessary for an inflammatory response. However, clusters 1 and 5 both had a lactate level greater than 4 mmol/L , which is a threshold that has been previously used in the early detection of potential sepsis cases [15]. Furthermore, only clusters 1 and 5 had a

majority count of mortality, which may follow lactate the result that high and intermediate levels of lactate may predict mortality in severe sepsis patients [16]. Interestingly, in cluster 3, none of the test values exceeded any of the critical thresholds, yet the most deaths and subjects belonged to this cluster. This suggests that some other clinical tests may be necessary for fine tuning the classification, or that the database needs to be processed in another manner – possibly considering the time scale.

Consequently, although clusters were formed, the Silhouette values for the clusters were not remarkable, since only four out five clusters had a value less than 0.25 and only cluster 2 was close to the 0.5 level. This demonstrates that the data used may need further processing or more patients are necessary to improve the structure confidence.

TABLE I. Summary of Clustering

Cluster #	# of Subjects	Sepsis Occurrence		Mortality		Silhouette width
		No	Yes	Alive	Dead	
1	12	11	1	3	9	0.126
2	12	11	1	5	7	0.408
3	881	697	184	682	199	0.183
4	157	139	18	81	76	0.104
5	14	7	7	3	11	0.136

TABLE II. Summary of clinical tests per cluster using the median value

Cluster #	# of Subjects	LAC (mmol/L)	WBC (# cells in thousands / mm ³)	RR (bpm)	TEMP (F)
1	12	8.36	4.47	23.3	96.23
2	12	1.97	11.71	43.27	98.47
3	881	1.89	9.67	18.95	98.56
4	157	1.93	19.00	18.29	98.23
5	14	10.88	20.57	24.75	99.51

However, there is potential for the clustering results to provide features that may be used for inference in a Bayesian network, since some of the clusters were separated based on some of the SIRS criteria and on the lactate levels.

To better understand the features that may be contained in the clustered groups, it would be meaningful to look at the probabilistic relationships demonstrated in the Bayesian

networks. Fig. 3.2a shows the Bayesian network (BN1) learned from temperature, RR, WBC, lactate, LOS, and SO. Fig. 3.2b shows the Bayesian network (BN2) learned from temperature, RR, WBC, lactate, LOS, SO, MAP. In a visual comparison between the two networks, BN2 lost an arc going from WBC to RR, and lactate to RR, but gained an arc going from lactate to MAP, and MAP to RR. Essentially, in BN2 the lactate node has an indirect path to RR through MAP, instead of direct one as in BN1. These relationships reflect the initial clustering results, especially in clusters 2 and 5 where the WBC levels and RR values are high. Also, the importance of lactate is shown in effecting the SIRS criteria measurements, which demonstrate that lactate levels are a key feature in sepsis detection.

With the arcs produced in BN1 and BN2, it is important to assess these arcs in an unbiased manner to choose the ideal learning scheme. Table III provides the goodness of fit estimates from the 10 fold cross-validation computation for each of the scoring criterion. In BN1, the model learned with the BGE scoring criterion was chosen since it had the lowest loss function value. For BN2, BIC (Fig. 3.2b) and BGE learned the same network, however BIC had a lower loss function value, therefore the final network was chosen using BIC. The overall goodness of fit estimates for BN1 are lower than BN2.

TABLE III. The goodness of fit for each BN

Scoring Criterion	Loss Function Value	
	BN1	BN2
BGE	15.413	19.95
BIC	15.433	19.93
AIC	15.473	19.94
LOG	15.423	19.97

Table IV shows the strength of the arcs in BN1 for the BIC criterion, and Table V shows the strength of the arcs in BN2 for the BIC and BGE criterion, where strength of 1 is the maximum and 0 is the minimum value possible. For BN2, all of the BGE arc strengths were equal or greater than the BIC arc strengths. Furthermore, in a comparison of the strengths of the common arcs in BN1 and BN2, BN1 had the

Dendrogram of hierarchical clustering on patients

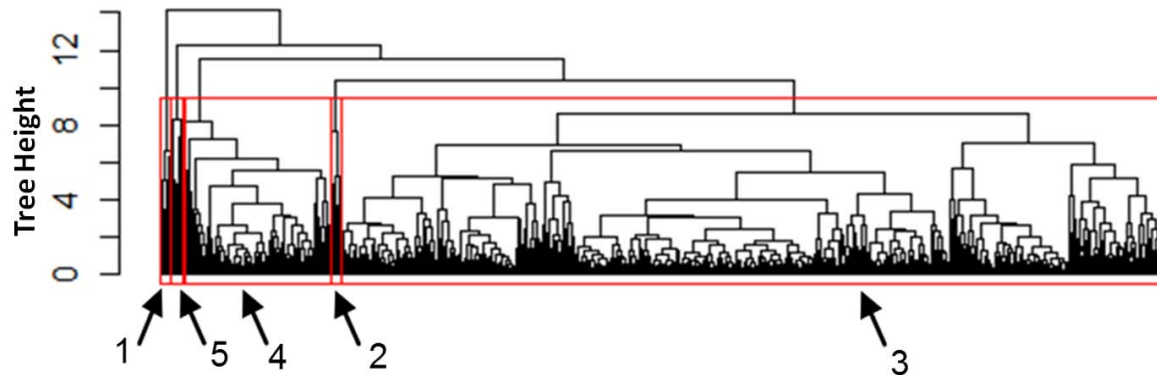


Figure 3.1: The dendrogram of the hierarchical cluster of septic patients with the inclusion of mortality. The red squares and arrows indicate the five major clusters.

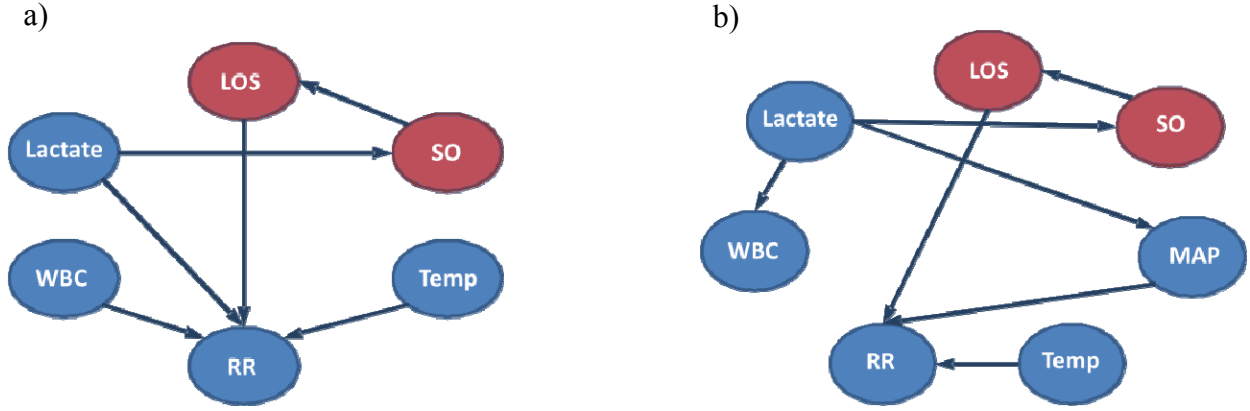


Figure 3.2: Bayesian Network 1 and 2, (a) and (b), respectively. (a) BN1 is constructed using BGE with the nodes temperature (Temp), RR, WBC, lactate (LAC), length of stay (LOS), and SO nodes. (b) BN2 constructed by BIC and BGE, using the same nodes as BN1 with the addition of MAP.

stronger arcs compared to the BIC learned BN2. However, when the BN1 is compared with the BGE learned network BN2, BN2 has the stronger arcs going from lactate to WBC and SO to LOS, whereas BN1 has stronger arcs for going from lactate to SO and LOS to RR.

TABLE IV. The strength of each arc in BN1

Arc Direction		Strength
From	To	BGE
Lactate	WBC	0.935
Lactate	RR	0.950
Lactate	SO	0.775
SO	LOS	0.710
TP	RR	1.00
LOS	RR	0.990
WBC	RR	0.770

TABLE V. The strength of each arc in BN2

Arc Direction		Strength	
From	To	BGE	BIC
Lactate	WBC	0.965	0.895
Lactate	SO	0.645	0.590
Lactate	MAP	1.00	1.00
SO	LOS	0.795	0.570
TP	RR	1.00	1.00
LOS	RR	0.945	0.945
MAP	RR	0.995	0.990

IV. DISCUSSION

In both BN1 and BN2 there is an arc going from lactate to sepsis occurrence, which is the only arc pointing to SO. Also, there is no arc pointing to lactate. This indicates that the lactate levels may be the most important factor in determining cases of sepsis. This reflects the use of lactate levels in early detection of septic patients for goal directed therapy [15] and in optimizing resuscitation of septic shock patients [7]. BN1 shows arcs pointing from lactate to WBC (as does BN2) and RR. However, lactate targets RR indirectly through MAP in BN2. This may suggest that

lactate may play an important role in both of these SIRS criteria.

Interestingly, BN2 shows that lactate levels may affect MAP measurements, which was shown in [8] to be independently associated with mortality in septic shock. However, in our model MAP is not pointing to sepsis, but this may be due to the fact that averaged patient data was used and that an analysis at different time scales may be necessary as in [8]. In addition, sepsis occurrence was pointing to LOS in both BN1 and BN2, but the strength of this arc is weaker compared to the other arcs.

One limitation of this study is that in producing BN2, the number of patients was reduced from 1076 to 741, which may have caused the strength of some arcs to decrease, especially of the one going from lactate level to sepsis occurrence. Another limitation is that the test measurements were averaged over time, which may have eliminated some important time dependent features in the clinical tests.

For this reason, in future analyses, it would be imperative to separate data into two time bins of 6 hours, and 48 hours or 72 hours as previously used in [7, 8, 15]. Further modeling of the networking structure would include incorporating mortality, as it has been shown that lactate levels have been able to predict mortality regardless of the presence of organ dysfunction [16].

In conclusion, it was shown that Bayesian networks of sepsis patients were able to capture the relationship between lactate levels and sepsis. Furthermore, it was shown that lactate levels may be predictive of the SIRS criteria. Future work incorporating time dependency of the clinical tests may provide useful insight into the relationship of MAP with sepsis occurrence. Nonetheless, it can be surmised that Bayesian networks of sepsis patients hold the promise of potentially evolving into a full scale clinical decision support system.

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